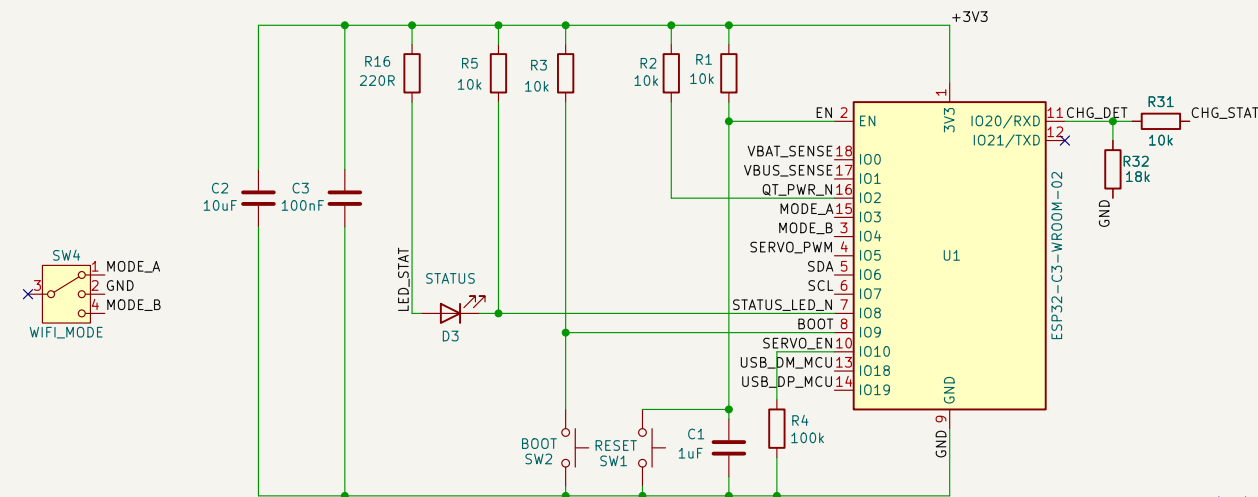


MCU – ESP32–C3–WROOM–02



Power Flags

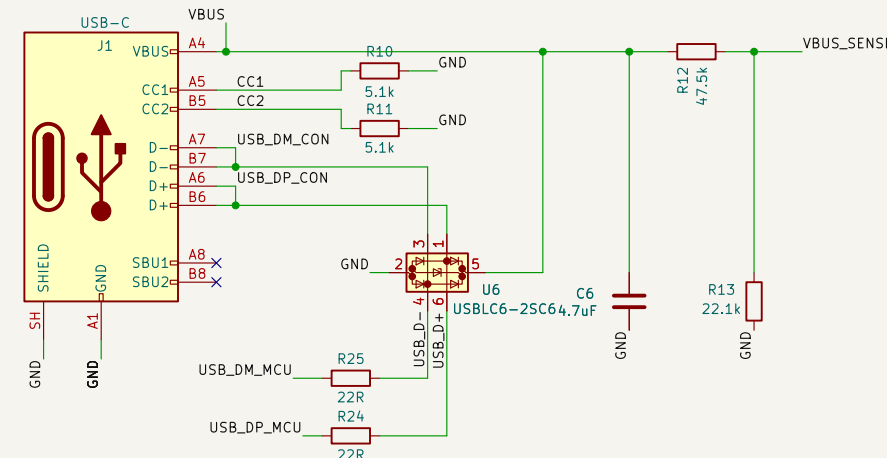


ESP32–C3 notes (MCU box):
 BOOT (SW2) low at reset -> SPI boot.
 I08 drives the STATUS LED active-low (LED off = strap high).
 I010 = SERVO_EN; R4 100k keeps the servo rail off at reset/sleep.
 EN: R1 10k + C1 1uF RC delay (Espressif HW design guide).
 USB: I018/I019 native USB Serial/JTAG – no UART bridge.

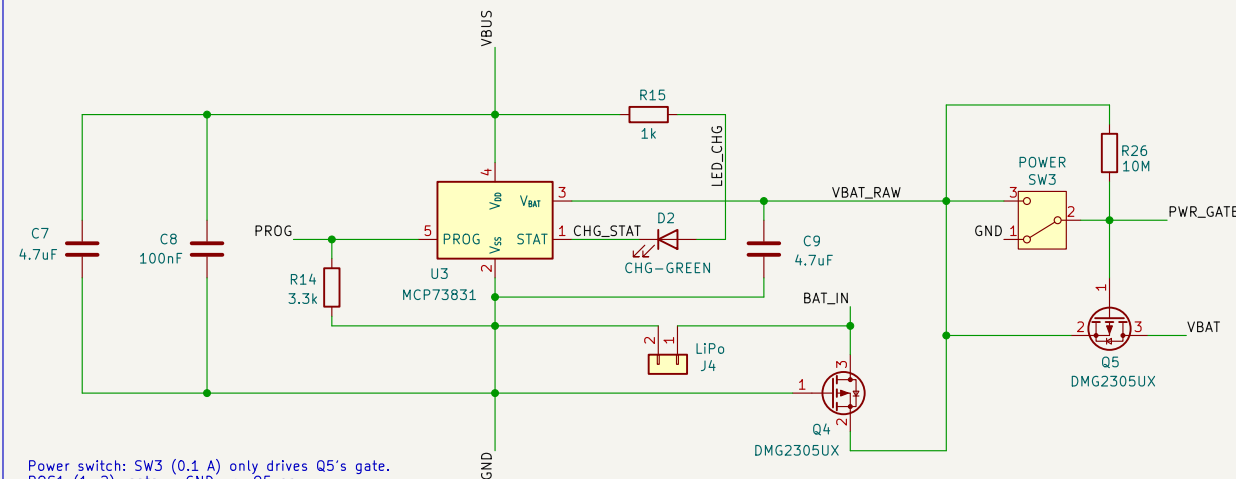
SW4 (OS103011MA7QP1, common = pin 2):
 POS1 MODE_A (IO3) low = OFF | POS2 both high = SPARSE | POS3 MODE_B (IO4) low = CONTINUOUS.

Internal pull-ups only. IO3/IO4 are deep-sleep wake-capable (GPIO0–5). VBAT_SENSE IO0 / VBUS_SENSE IO1 = ADC1.

USB–C Input + USB data + ESD + VBUS sense

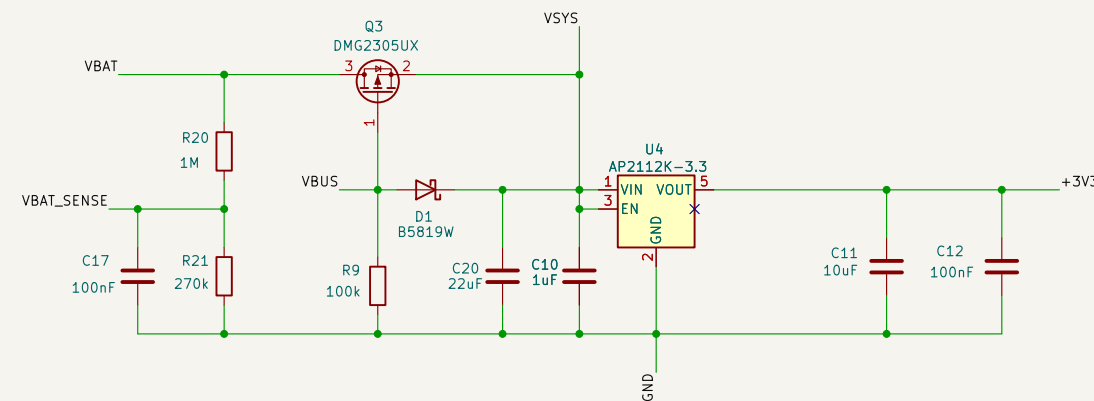


LiPo Charger (MCP73831) + power switch

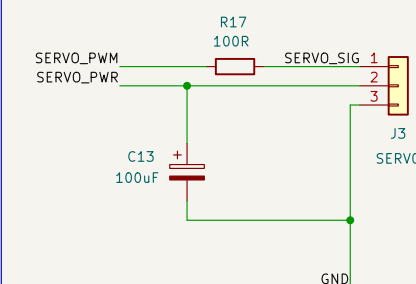


Power switch: SW3 (0.1 A) only drives Q5's gate.
 POS1 (1–2): gate = GND -> Q5 on
 POS2 (2–3): gate = source -> Q5 off
 R26 10M holds Q5 off in the break-before-make gap.

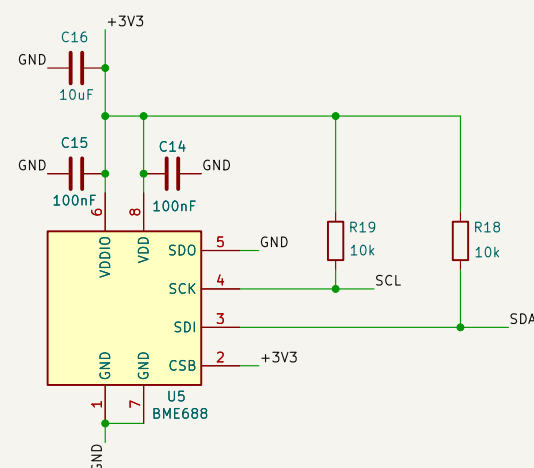
Power Path + 3V3 LDO



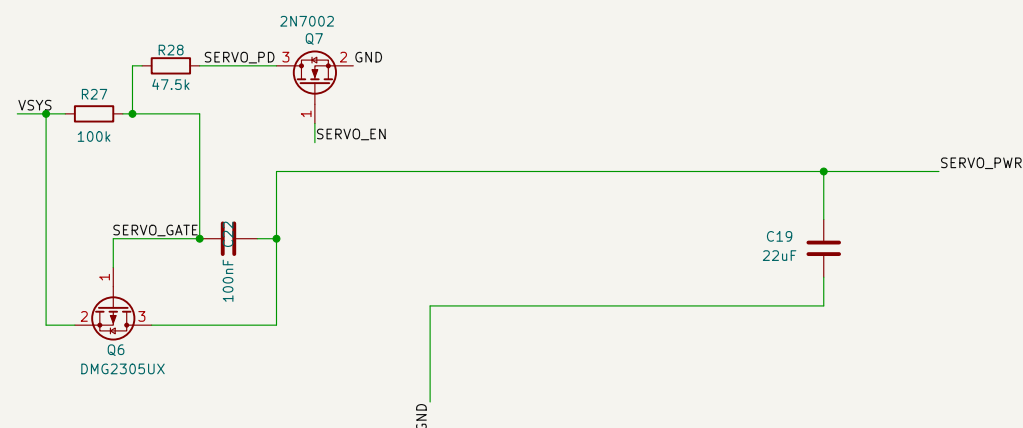
Servo Output



Air Sensor (BME688)

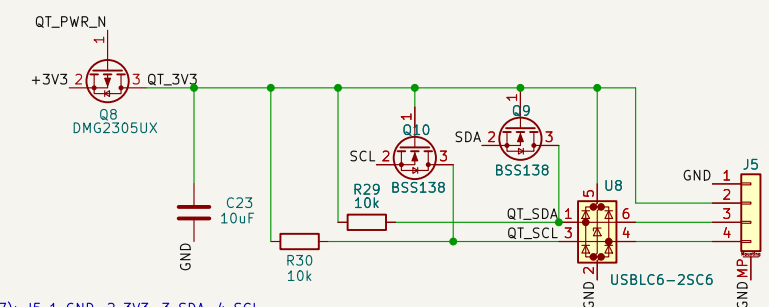


Servo Rail Load Switch



Servo rail load switch (v8: no boost, the servo runs from VSYS = USB via D1 ~4.5 V or the battery 3.0–4.2 V).
 Q6 connects SERVO_PWR to VSYS only while SERVO_EN (IO10) is high.
 Q7 pulls the gate to -VSYS–2.5 V via R28; R4 keeps it off at reset/sleep.
 C22 (gate-drain) slews SERVO_PWR at ~0.5 V/ms -> ~60 mA inrush into C13 + C19 (122 uF).
 Firmware: SERVO_EN high >= 50 ms before a move; skip moves below VBAT ~3.5 V.

Stemma QT



STEMMA QT port (v7): J5 1 GND, 2 3V3, 3 SDA, 4 SCL.
 Q8 switches it from IO2 (low = on; R2 keeps it off at reset/sleep).
 Q9/Q10 isolate the bus when off. R29/R30 port pull-ups, U8 ESD.
 BME688 now at 0x76 (SDO = GND) – 0x77 free for add-ons.
 R31/R32: charger STAT (5 V) -> IO20 CHG_DET. Low = charging (with VBUS), high = done.

ESP32–C3 + BME688 + SG92R (v7)

Sheet: /
 File: air-quality-pcb-v8.kicad_sch

Title: Air Quality Checker

Size: A3

Date:

Rev: H

KiCad E.D.A. 10.0.6

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